

THE HIMALAYAS

Formation • Divisions • Peaks • Purvanchal

Fold Mountains formed by **Continent–Continent Convergence** between the Indian & Eurasian Plates

- Headings & Definitions ● Explanations ● Highlighter = Key Facts 🎯 Exam Tip ✗ Common Mistake
🧠 Memory Trick

StudyPlanner



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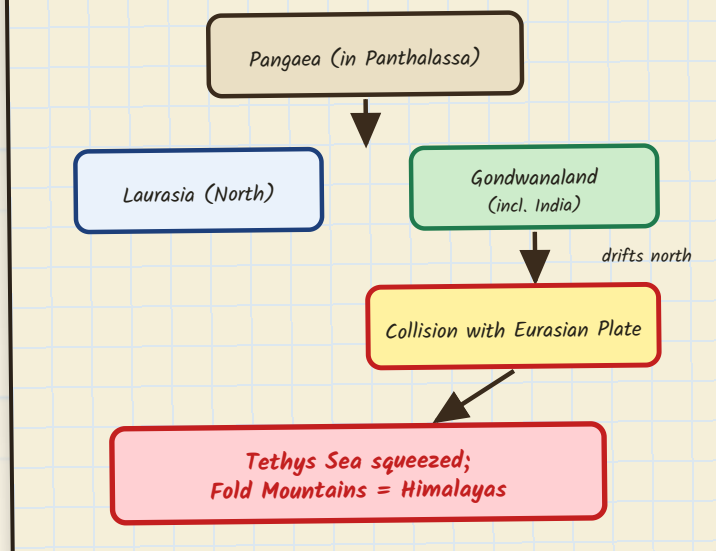


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HOW THE HIMALAYAS WERE FORMED

About 150 million years ago, the supercontinent **Pangaea** (surrounded by the ocean **Panthalassa**) broke into two: **Laurasia** on top and **Gondwanaland** on the bottom. India was part of **Gondwanaland**. India then drifted **upward** toward the **Eurasian Plate**.

🧠 Fig 1: Formation of the Himalayas



Both the Indian Plate and the Eurasian Plate are **continental plates** of similar density — so neither was willing to sink under the other. This is called **continent–continent convergence**, and it produced fold mountains: the Himalayas.

The Tethys Sea – The Evidence

Between the two colliding plates lay a sea called the **Tethys Sea**. We know it existed because **marine fossils** have been found high up in the mountains — sea sediments don't get there unless a sea once existed in that spot. The Himalayas were **not** formed by a sudden collapse; they built up slowly, from sea-bed to peak, over millions of years.

Exam Tip

The presence of **marine fossils** in the high Himalayas is the classic evidence question — always link it to the vanished **Tethys Sea**.

Common Mistake

Don't picture the Himalayas as forming in one dramatic "collapse" — the process was gradual, and it's still ongoing: the range's height increases by roughly 5 cm every year.



Memory Trick

Pangaea → splits → Laurasia (north) + Gondwanaland (south, with India) → India drifts up → hits Eurasia → Tethys Sea squeezed out → Himalayas rise.

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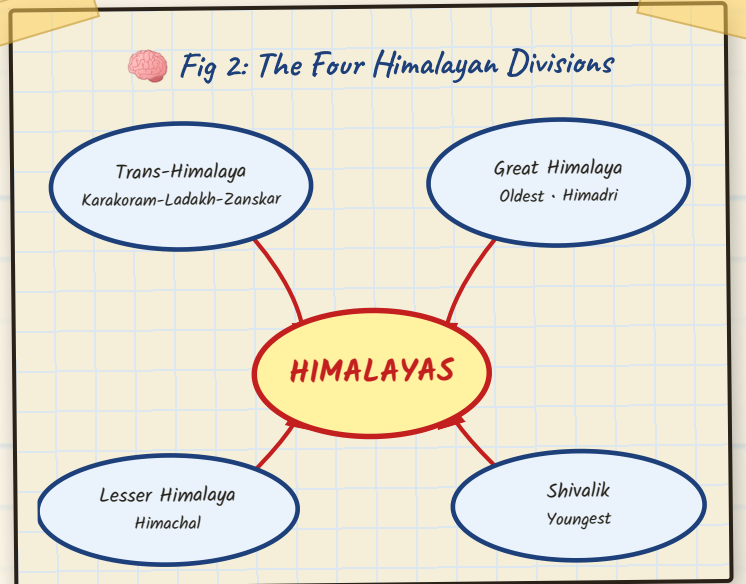
THE FOUR DIVISIONS + THE TRANS-HIMALAYA

From north to south, the Himalayas are divided into four ranges: the **Trans-Himalaya**, the **Great Himalaya**, the **Lesser Himalaya**, and the **Shivalik**.

Division	Other Names	Notes
Trans-Himalaya	—	Believed to have formed from Tethys Sea deposits ; sometimes not even counted as part of the Himalaya proper

Great Himalaya	Inner Himalaya, Himadri	The oldest range; core made of granitic rock; average peak height 6000 m
Lesser Himalaya	Himachal, Middle Himalaya	Average height 4000 m
Shivalik	Outer Himalaya	The youngest range

Trans-Himalayan Ranges (In Order)



Karakoram Glaciers

In the Karakoram range, remember these glaciers: **Hispar, Baltoro, Biafo, and Siachen.**

Memory Trick

Just remember "**H-B-B-S**" for the Karakoram glaciers: Hispar, Baltoro, Biafo, Siachen.

In **1984**, **Operation Meghdoot** took place at the Siachen Glacier. Siachen is the **second-longest non-polar glacier** in the world — the longest is the **Fedchenko Glacier** in Tajikistan.

Common Mistake

Don't say Siachen is the **longest** non-polar glacier — it's the **second-longest**; the Fedchenko Glacier in Tajikistan holds the top spot.



Exam Tip

A "mountain range" simply means a whole connected series of mountains — keep this basic definition ready for direct-definition questions.

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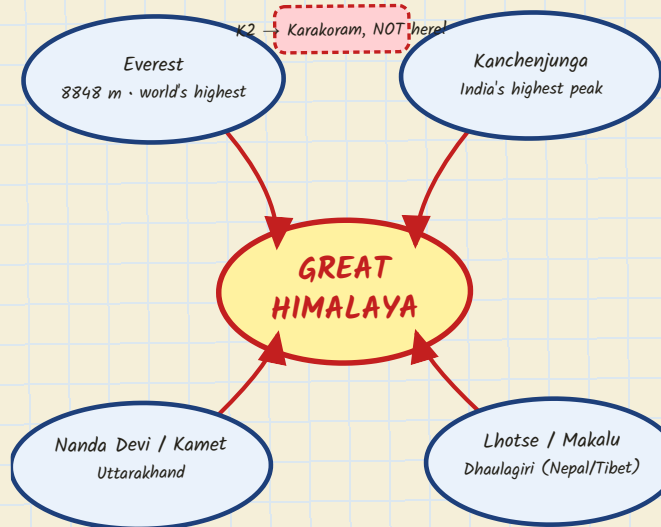
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THE GREAT HIMALAYA (HIMADRI)

The Great Himalaya is the **oldest** of the four ranges, also called the **Inner Himalaya** or **Himadri**. This is where the **world's highest peaks** are found. Its core is made of **granitic rock**, and the average height of its peaks is **6000 m**. Its western-most point is **Nanga Parbat**; the range's eastern-most extent reaches into **Arunachal Pradesh**.

Fig 3: Peaks of the Great Himalaya



Peak	Location	Height	Notes
Mount Everest	Nepal / Tibet border	8848 m	World's highest; Sagarmatha (Nepal), Chomolungma (Tibet)
Kanchenjunga	Sikkim, India	~8586 m	India's highest peak; world's 3rd highest

Mount Lhotse	Tibet	8516 m	World's 4th highest peak
Mount Makalu	Nepal	—	Among the world's highest peaks
Dhaulagiri	Nepal	—	Among the world's highest peaks
Nanda Devi	Uttarakhand, India	7816 m	Highest peak in Uttarakhand
Kamet	Uttarakhand, India	—	Also in the Great Himalaya

 Wait — isn't K2 India's highest peak? (tap to expand) ▼

✗ Common Mistake

Don't club K2 with the Great Himalaya's top peaks — it's a **Karakoram** (Trans-Himalaya) peak, in a disputed PoK location, not part of the Great Himalaya range at all.

🎯 Exam Tip

Order to remember: Everest (world #1) → K2 (world #2, Karakoram) → Kanchenjunga (world #3, India's highest) → Lhotse (world #4).

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LESSER HIMALAYA, SHIVALIK + THE VALLEYS

Lesser Himalaya (Himachal) — Regional Names

The Lesser (Middle) Himalaya has an average height of 4000 m and is known by different local names across regions:

Region	Local Name of the Range
Kashmir	Pir Panjal Range
Himachal Pradesh	Dhauladhar Range

Uttarakhand

Nag Tibba Range

Nepal

Mahabharat Range

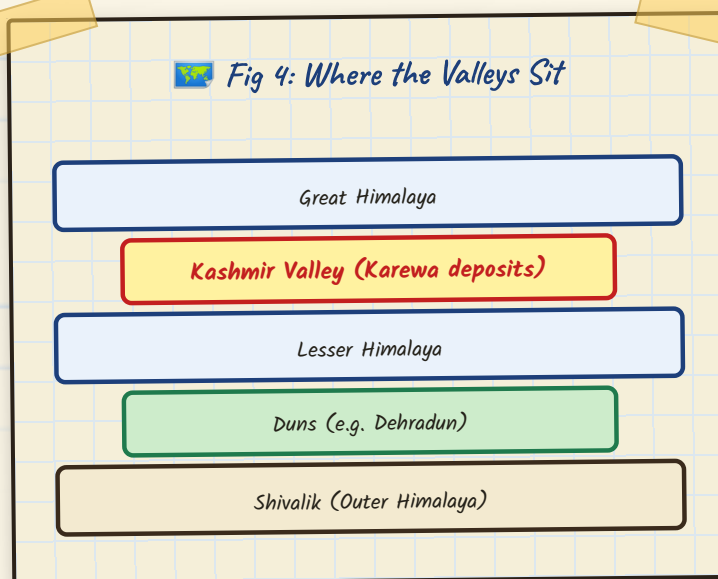
✗ Common Mistake

Don't confuse **Dhauladhar** (the Lesser Himalaya range in Himachal Pradesh) with **Dhaulagiri** (a Great Himalaya peak in Nepal) — similar-sounding names, completely different things!

Shivalik (Outer Himalaya)

The Shivalik range is the youngest of the four divisions. It disappears in the Northeast, where it's replaced by the **Duars** — soft, unconsolidated deposits.

 Fig 4: Where the Valleys Sit



Kashmir Valley & Karewa

★ The **Kashmir Valley** lies between the **Great Himalaya** and the **Lesser Himalaya**. Here you find Karewa — soft, unconsolidated glacial/lacustrine deposits — which are especially helpful for growing **saffron** (a local variety).

Duns

Duns are longitudinal valleys found between the Lesser Himalaya and the Shivalik. The largest Dun is Dehradun; another named Dun is **Patli Dun**.

Uttarakhand's Grasslands & Tribes

Summer grasslands in Uttarakhand are called **Bugyal** — home to the **Bhotia** tribe. The **Gujjar** and **Bakarwal** tribes are also found here.



Exam Tip

Keep the valley-location pairs clear: **Karewa** → **Kashmir Valley** (between Great & Lesser Himalaya); **Duns** → **between Lesser Himalaya & Shivalik**.



Memory Trick

Karewa = Kashmir's saffron soil; **Duns** = Dehradun's valley soil — both are soft, unconsolidated deposits, just in different locations!

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REGIONAL (LONGITUDINAL) DIVISIONS + PURVANCHAL

The Himalayas are also divided **longitudinally**, based on the rivers that mark their boundaries:

Between Rivers	Himalaya Name
Indus – Sutlej	Punjab Himalaya
Sutlej – Kali	Kumaon Himalaya
Kali – Teesta	Garhwal / Nepal Himalaya
Teesta – Brahmaputra	Assam Himalaya



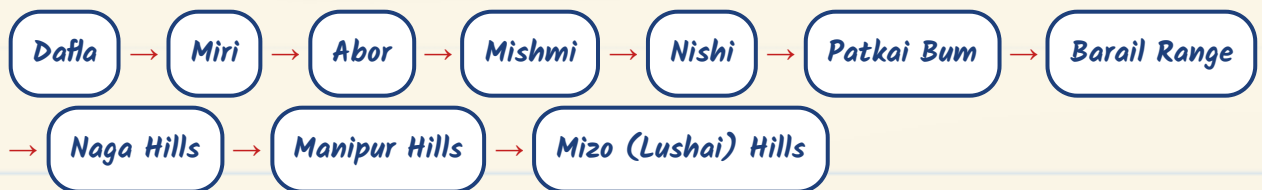
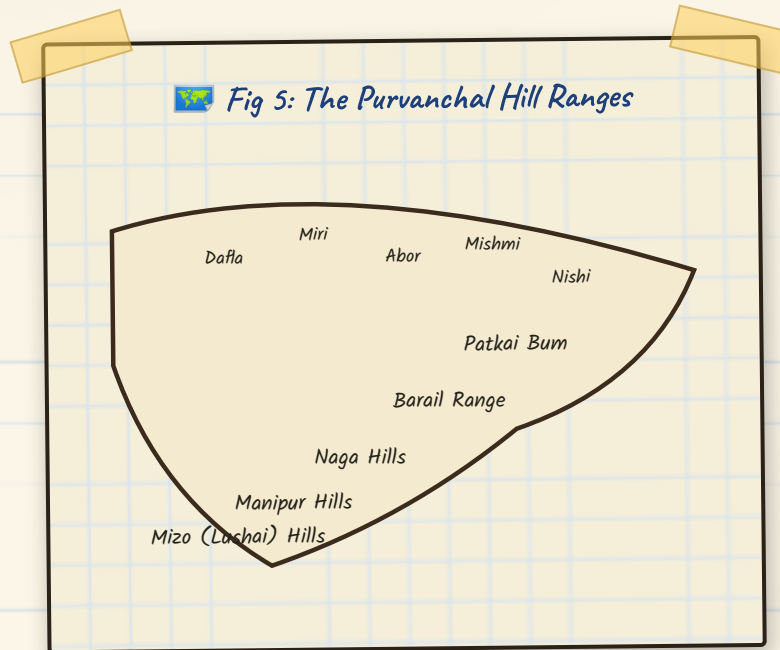
Memory Trick

Think "IPS" — **Indus** → **Punjab Himalaya** → **Sutlej** — to lock in the first segment, then move river by river: **Sutlej–Kali** (Kumaon), **Kali–Teesta** (Garhwal/Nepal), **Teesta–Brahmaputra** (Assam).

Focus on *which Himalaya lies between which two rivers* — exact locations aren't usually asked, the river-pairing is what matters.

Purvanchal Himalaya (Northeast)

Beyond the Assam Himalaya, in the Northeast, lie the Purvanchal Himalayas — a set of hill ranges across Arunachal Pradesh and the northeastern states:



"Mizo Lushai is the same" — Mizo Hills are also known as the Lushai Hills.

Exam Tip

Write the Purvanchal chain out by hand a few times — **Dafla, Miri, Abor, Mishmi, Nishi, Patkai Bum, Barail, Naga, Manipur, Mizo (Lushai)** — repetition is the only trick that works here.

Common Mistake

Don't drop the alternate name — **Mizo Hills = Lushai Hills**; exams may use either name for the same range.

★ Exam-Ready Summary

- **Formation:** Continent–continent convergence between the Indian & Eurasian plates; Tethys Sea evidence via marine fossils; still rising ~5 cm/year
- **Four divisions (oldest→youngest):** Great Himalaya → Lesser Himalaya → Shivalik (Trans-Himalaya sits apart, from Tethys deposits)
- **Trans-Himalaya order:** Karakoram → Ladakh → Zaskar; glaciers: Hispar, Baltoro, Biafo, Siachen
- **Great Himalaya:** Himadri/Inner Himalaya; Everest, Kanchenjunga, Lhotse, Makalu, Dhaulagiri, Nanda Devi, Kamet (K2 excluded — it's Karakoram)
- **Lesser Himalaya names:** Pir Panjal (Kashmir), Dhauladhar (HP), Nag Tibba (Uttarakhand), Mahabharat (Nepal)
- **Valleys:** Kashmir Valley (Karewa, saffron) between Great & Lesser Himalaya; Duns (Dehradun) between Lesser Himalaya & Shivalik
- **Longitudinal divisions:** Punjab, Kumaon, Garhwal/Nepal, Assam Himalaya — plus Purvanchal in the Northeast

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 Print Notes

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